MILITARY SPECIFICATION

BALLISTICS COMPUTER SUBASSEMBLY: 8635274
(ZEROING CONTROL UNIT)

This amendment forms a part of Military Specification MIL-B-60120(MU), dated 5 June 1964.

Paragraph 2.1, Specifications: Delete "MIL-I-45208 - Inspection System Requirements".

Paragraph 3.4.1: Delete last sentence.

Paragraph 3.4.3: Delete and substitute:

"3.4.3 Sealing.- When pressurized to an internal pressure of 5 ± 0.1 psig, the zeroing control unit shall not exhibit a pressure drop greater than 0.2 psig for a period of at least 2 minutes."

Add the following paragraph:

"3.4.4 Storage temperature.- The unit shall show no evidence of physical failure and shall meet all requirements of this specification at standard ambient temperature of 60° to 90°F after having been exposed to and thermally stabilized at ambient temperature of -80°F and +160°F."

* Paragraph 3.6.1, TABLE I: Delete and substitute:

"TABLE I. ZEROING CONTROL OUTPUT VOLTAGE RATIOS

| Ratio Setting | Zeroing Knob Setting | Tolerance |
|---|---|---|
| .100000 | Up or Right 3 mils | ± 0.1 Mil |
| .233333 | Up or Right 2 mils | ± 0.1 Mil |
| .366666 | Up or Right 1 mil | ± 0.1 Mil |
| .500000 | Zeroed | ± 0.05 Mil |
| .633333 | Down or Left 1 mil | ± 0.1 Mil |
| .766666 | Down or Left 2 mils | ± 0.1 Mil |
| .900000 | Down or Left 3 mils | ± 0.1 Mil" |

Paragraph 3.6.3: Delete "20 amperes" and substitute "20 plus 0.20 minus 0.0 amperes"; Delete "10 plus 0.0 minus 0.20 amperes" and substitute "15 plus 0.0 minus 0.20 amperes".



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Paragraph 3.7.1: Delete and substitute:

"3.7.1 Boresight-normal knob operation force.- The maximum force required to pull the boresight-normal knob out from the normal position to the boresight position shall not exceed 18 pounds. The maximum force required to push the boresight-normal knob in from the boresight position to the normal position shall not exceed 18 pounds."

Paragraph 4.1.1: Delete and substitute:

"4.1.1 General provisions.- The component and subassembly inspection requirements of MIL-F-13926 form a part of the quality assurance provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109."

Add new paragraph:

"4.1.2 First article (initial production) inspection.- When required by the contract, an initial production sample shall be submitted to a Government approved facility in accordance with the contract schedule for test and evaluation. The sample shall consist of three zeroing control units plus three of every component and subassembly covered by Supplementary Quality Assurance Provisions (SQAP). The sample shall be manufactured in the same manner, using the same materials, equipment, processes, and procedures, as used in regular production. All parts and materials, including packaging and packing, shall be the same as used for regular production and shall be obtained from the same source of supply. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to all requirements of the contract and submit a record of this inspection with the sample, including certificates of conformance for materials. After inspection by the contractor and provisional acceptance at source by the Government, the sample shall be inspected at a Government facility or such other facility as specified in the contract, for all the requirements of the drawings and specifications."

Paragraph 4.2.2, Table II: Add the following characteristic:

| | | | |
|---|---|---|---|
| "108. | Sealing | 3.4.3 | 4.5.13 |
| 109. | Dielectric strength | 3.5.2 | 4.5.3 |
| 110. | Insulation resistance | 3.5.3 | 4.5.4" |

Paragraph 4.3: Delete "Control tests" and substitute "Special sampling".

Paragraph 4.3.1, last sentence: Delete "control"; In tabulation heading delete "Control test" and substitute "Characteristic".



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Paragraph 4.3.2: Delete second sentence and substitute; "Each sample shall meet the following requirements and tests, and shall then meet the requirements and tests in Table II."; Delete tabulation and substitute:

| "Characteristic | | Requirement | Test Procedure |
|---|---|---|---|
| 304. | Storage temperature (-80° and +160°F) | 3.4.4 | 4.5.1.4 |
| 305. | Output voltage | 3.4.1, 3.6.1 | 4.5.1.1, 4.5.5 |
| 306. | Switch resistance | 3.4.1, 3.6.2 | 4.5.1.1, 4.5.6 |
| 307. | Circuit breaker function | 3.4.1, 3.6.3 | 4.5.1.1, 4.5.7 |
| 308. | Boresight knob operating force | 3.4.1, 3.7.1 | 4.5.1.1, 4.5.8 |
| 309. | Zeroing knob operation | 3.4.1, 3.7.2 | 4.5.1.1, 4.5.9 |

NOTE: The tests for Characteristics 305 through 309 inclusive shall be conducted at -40° and +150°F."

Paragraph 4.3.3: Delete and substitute:

"4.3.3 Failure of sample.- Should any one item of a special sampling fail to meet the specified test requirements, acceptance of the represented inspection lot shall be suspended by the Government until necessary corrections have been made by the contractor and resubmitted items have been approved (see 4.2.3)."

Paragraph 4.4, first sentence: Delete "Measuring and Test Equipment requirement of MIL-I-45208 and all".

Paragraph 4.4.2, Item Number 2 description: Delete:

"1 megohm ± 10% and substitute "1 megohm minimum"; Item 3 Description: Delete and substitute "0.5 volts rms, 400 cps ± 1%, 2% maximum harmonic distortion"; Item Number 4 description: Delete and substitute "0-30 volts, maximum ripple - 0.10 volt peak-to-peak, 30 amperes output".

Paragraph 4.5.1.2: Delete last sentence and substitute:

"The zeroing control unit shall then be subjected to the tests specified in Table II to determine conformance with the applicable requirements referenced therein."

Paragraph 4.5.1.3: Delete and substitute:

"4.5.1.3 Sealing.- The pressure test equipment utilized for this test shall be capable of internally pressurizing the zeroing control unit



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to the psig value specified. The pressure gauge of the test equipment shall permit measuring any variation in the internal pressure in increments of 0.1 psig. Remove the Plug MS49005-1 from the side of the zeroing control unit and replace it with an air valve containing a spring loaded valve stem. The pressure shall be applied to the zeroing control unit through this valve. With the cover of the zeroing control unit secured in a closed position, pressurize the zeroing control unit with clean dry air to the psig value specified in 3.4.3. The applied pressure shall be retained for a minimum of 2 minutes. Any evidence of leakage in the zeroing control unit at the end of the 2 minute time period shall not exceed the tolerance specified in 3.4.3, as measured on the pressure gauge of the testing equipment. Upon completion of this test, remove the air valve and replace the Plug MS49005-1 in the zeroing control unit."

Add the following paragraph:

"4.5.1.4 Storage temperature.- The testing equipment utilized in this test shall be in accordance with the "Test Facilities" requirements of MIL-F-13926 and the conditions of 3.4.4. The zeroing control unit shall be exposed to each temperature extreme specified in 3.4.4 for a minimum of 5 hours. After each temperature exposure the zeroing control unit shall be stabilized in an ambient temperature between +60° and +90°F and subjected to a visual, tactile and audible examination for evidence of physical failure. The zeroing control unit shall then be subjected to the tests in 4.5.1.1 through 4.5.9 inclusive to determine conformance to 3.4.4. Caution shall be exercised during the temperature cycling to avoid subjecting the zeroing control unit to thermal shock."

Paragraph 4.5.6: After "small c and large F," delete "small q and large F" and substitute "small g and large F".

Paragraph 4.5.7: Delete "10 plus 0.0 minus 0.20 amperes" and substitute "15 plus 0.0 minus 0.20 amperes". Delete "20 amperes" and substitute "20 plus 0.20 minus 0.0 amperes".

Paragraph 4.5.8, last sentence: Delete "12 pounds" and substitute "18 pounds".

*The outside margins have been marked to indicate where additions to the previous amendment have been made. This has been done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in this notation. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content as written irrespective of the marginal notation and relationship to the last previous amendment.

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